

AKHILESH KUMAR MISHRA

Full Stack Engineer (MERN) | React · Node.js · MongoDB — Payments at 2M+ Users

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PROFILE

500,000+ payment transactions a day for 2 million+ users on India's national e-governance platform. Cut p95 read latency 40% by rewriting the MongoDB aggregation pipelines behind it, and held AEPS and MATM settlement at a 99% success rate by making the payment path idempotent and self-reconciling. React, TypeScript and Node.js end to end — schema design through Dockerised CI/CD.

CORE TECHNOLOGY

Frontend: React.js, Next.js SSR/ISR, TypeScript, JavaScript (ES6+), Redux Toolkit, React Query, Tailwind CSS, Material UI, code splitting, Core Web Vitals

Backend & APIs: Node.js, Express.js, REST API design, OpenAPI/Swagger, WebSockets, JWT, OAuth 2.0, RBAC, rate limiting, idempotency keys, API versioning

Data & Caching: MongoDB, aggregation pipelines, compound indexing, sharding, Mongoose, Redis, PostgreSQL, query profiling, schema design

Testing, Cloud & Practice: Jest, React Testing Library, Supertest, Docker, GitHub Actions CI/CD, AWS EC2/S3/CloudFront, Nginx, Grafana, Sentry, Agile/Scrum, code review

EXPERIENCE

Software Development Engineer — *CSC e-Governance Services India Ltd. (MeitY, Government of India)*

Ministry of Electronics & Information Technology, Govt. of India

12/2023 – PRESENT | New Delhi, India

- Held AEPS and MATM settlement at a **99% success rate** through peak load, by making the UPI and XML payment path idempotent — request keys, exponential-backoff retries and an automated end-of-day reconciliation job.
- Cut **p95 read latency 40%** on the transaction APIs serving **2 million+ users**, by collapsing N+1 **Mongoose** queries into a single **MongoDB** aggregation pipeline and compound-indexing the hot collections.
- Streamed **500,000+ daily transactions** to operations in real time, by building the transaction and geolocation dashboards in **React.js** and **Redux Toolkit** over a **WebSocket** feed with memoised selectors.
- Contained a bad release to one service instead of the whole portal for **2 million+ users**, by decomposing a monolithic e-governance system into **Node.js** and **Express.js** microservices behind a versioned REST gateway.
- Blocked **99.9%** of unauthorized access attempts on the payments API, by adding **JWT** access/refresh rotation, **Redis**-backed sessions, device binding, RBAC middleware and per-device OTP rate limiting.
- Retired manual releases across all **3** payment platforms, by containerising the services with **Docker** and gating every merge on **GitHub Actions** running **Jest** unit and **Supertest** integration suites.

Software Engineer (Contract) — *Independent Client Engagement*

Real-Time Telemetry & Geofencing Platform

06/2023 – 11/2023 | Remote, India

- Shipped a field-workforce platform used by **1,000+ agents** end to end — schema design through AWS deployment — on **React.js**, **Node.js**, **Express.js** and **MongoDB**.
- Brought dashboard load time down **30%**, by route-level code splitting of the **React.js** map view, virtualising marker rendering and streaming positions over **WebSockets** instead of polling.
- Removed **45%** of redundant API calls, by moving server state onto **React Query** with stale-while-revalidate caching and consolidating client state in **Redux Toolkit**.

SELECTED PROJECTS

Himalayan Edges — MERN Commerce PWA

React.js / Node.js / Express.js / MongoDB / PWA / AWS S3

Lifted page-load speed and client usage 30% at a 20% cart conversion rate, by shipping a service-worker offline cache, route-level code splitting and S3-backed asset delivery on a React, Express and MongoDB storefront.

DigiPay Web SDK & React Widgets

React.js / TypeScript / npm workspaces / Webpack / REST

Embeddable React component library and browser SDK published from an npm workspaces monorepo, giving merchant sites a drop-in payment UI over a versioned REST API. github.com/GitHub-akhilesh/Django_apis_digipay

Task Platform — React Web and React Native

React.js / React Native / Node.js / Docker / Webpack

React web client and React Native mobile app over a containerised REST backend, built as a workspace monorepo with Webpack builds and Docker Compose orchestration. github.com/GitHub-akhilesh/To-do-list-app

Real-Time Geofencing & Telemetry Dashboard

React.js / Node.js / Express.js / WebSockets / MongoDB / Redis

Delivers live GPS positions and geofence alerts to 1,000+ field agents with no polling anywhere in the path, by streaming over WebSockets from an Express ingestion API into time-indexed MongoDB collections.

EDUCATION

Bachelor of Technology (B.Tech.) | Kanpur Institute of Technology, Kanpur, UP

2019 - 2023

CERTIFICATIONS

IIT Kanpur — Cyber Security Certification

Advanced MERN Stack Developer Certification

Data Structures & Algorithms Certification

Database Systems & SQL Certification

ACHIEVEMENTS

- First Class Honors, B.Tech. — top academic rank in cohort.
- HackerRank 4-Star Gold badge in SQL.